Last name _____

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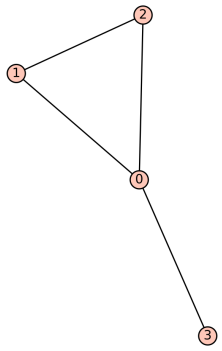
LARSON—MATH 756—CLASSROOM WORKSHEET 15
Knuth—Sandwich Theorem

Review

1. Show: for any graph, $\alpha \leq \vartheta$.
2. Find $\vartheta(K_{2,3})$.
3. Find $\vartheta(K_{m,n})$ $(n \geq m)$.

New

1. Find $\vartheta(G)$ for any bipartite graph $G = (A, B)$.



2. Find $\vartheta(G)$ for the paw graph.

3. Find $\vartheta(G)$ for any Kőnig-Egerváry graph G.

Knuth's Notation

4. What is unusual about Knuth's notation?

Knuth's Linear Algebra Background

5. What linear algebra results is Knuth referring to?

Characteristic Labelings

6. What is the *characteristic labeling* of a set $S \subseteq V(G)$?

7. What is a *stable labeling* of a stable set $S \subseteq V(G)$?

8. What is a *clique labeling* of a clique $S \subseteq V(G)$?

STAB and QSTAB

9. What is $\text{STAB}(G)$ for a graph G?

10. What is $\text{QSTAB}(G)$ for a graph G?

11. Why is $\text{STAB}(G) \subseteq \text{QSTAB}(G)$?